EECE 3624: Microprocessors

Fall 2026

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PreLab05

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## Prelab Questions

**Question 1.** State the function of each I/O port register: DDRx, PORTx, and PINx. Then, if DDRD2 = 0 and PORTD2 = 1, explain how PD2 is configured.

**Answer1:** DDRx: The DDRx controls the direction of the pins in PORTx
PORTx: The PORTx controls the value at the output by controlling the pull-up resistor
PINx: Reads the input at the pins.
If DDRD2= 0 and PORTD2=1, then PD2 is an input with the internal pull up resistor is enabled.

**Question 2.** Complete the joystick interrupt wiring below, and briefly explain why the external jumper wires are required.

**Answer2:** Joystick UP: PB3 → PD3 / INT3
Joystick DOWN: PB1 → PD1 / INT1
Why are the jumper wires required? – PORTB.1 and PORTB.3 cannot generate interrupts. The jumper wires physically connect the joystick to the external-interrupt pins.

**Question 3.** For external interrupts, state the purpose of EICRA, EIMSK, the SREG Ibit/ SEI, and the interrupt vector. Which one determines whether the interrupt is triggered by a rising or falling edge?

**Answer3.** EICRA: External Interrupt Control Register A; controls how the external interrupt is triggered
EIMSK: External Interrupt Mask Register; enables/disables specific e. interrupts.
SREG I-bit / SEI: Global Interrupt Enable bit
Interrupt vector: location in memory that tells the processor where to go once an interrupt is triggered
Rising/falling-edge selection is controlled by: EICRA

**Question 4:** The ISRs change BlinkFreq while the main program uses BlinkFreq to control the blink rate. Complete the logic below and state what should happen at the two limits.

**Answer4.** UP ISR: if BlinkFreq LESS THAN 15, then INCREMENT BlinkFreq
DOWN ISR: if BlinkFreq GRATER THAN 1, then DECREMENT BlinkFreq
If BlinkFreq = 15 and UP is selected: BlinkFreq does not change, stays at 15
If BlinkFreq = 1 and DOWN is selected: BlinkFreq does not change either, stays at 1

**Question5:** LED0–LED3 display BlinkFreq in 4-bit binary on PORTC.3:0 (PC3 = MSB, PC0 = LSB). Write the four output bits for each value.

**Answer5.** BlinkFreq = 1: PC3 PC2 PC1 PC0 = 0 0 0 1
BlinkFreq = 5: PC3 PC2 PC1 PC0 = 0 1 0 1
BlinkFreq = 10: PC3 PC2 PC1 PC0 = 1 0 1 0
BlinkFreq = 15: PC3 PC2 PC1 PC0 = 1 1 1 1

**Question6.** Assume BlinkFreq = 6. The user presses and releases the UP joystick. Number the following events in the order they occur.

1 A rising edge occurs on the interrupt pin.
2 The corresponding ISR begins.
3 BlinkFreq becomes 7.
4 LEDs 0–3 display 0111.
5 RETI returns control to the main program.
6 The main program uses the new blink rate.

Github link:

https://github.com/Jedelo90/EECE3624-Lab05-JoseD.git